Voiding Dysfunction

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 102	Surgical, Radiographic and Endoscopic Anatomy of the Male Pelvis
Ch 103	Physiology and Pharmacology of the Bladder and Urethra
Ch 104	Pathophysiology and Classification of LUT Dysfunction
Ch 106	Evaluation and Management of Males With Urinary Incontinence
Ch 107	Urodynamic and Videourodynamic Evaluation of the LUT
Ch 109	Neuromuscular Dysfunction of the Lower Urinary Tract
Ch 110	Overactive Bladder
Ch 111	Underactive Detrusor
Ch 113	Pharmacologic Management of Storage and Emptying Failure
Ch 115	Electrical Stimulation and Neuromodulation
★	Putting it together: voiding dysfunction decisions

102 Surgical, Radiographic and Endoscopic Anatomy of the Male Pelvis

1 · Bony pelvis

- Sacrum + paired innominate bones (**ilium, ischium, pubis** fused at the acetabulum); SI joints posteriorly, cartilaginous symphysis anteriorly.
- **Sacrotuberous** (→ ischial tuberosity) and **sacrospinous** (→ ischial spine) ligaments resist upward sacral movement; sacrospinous divides the sciatic foramen into **greater and lesser**.
- Pelvic brim **50–60°** to horizontal. **Arcuate line** (promontory → pubis) separates false from true pelvis. Iliac crest at **L4**.
- **Cooper's (pectineal) ligament**: anchors sutures in hernia repair and urethral suspension.

2 · Lower abdominal wall

Fascia and extravasation

- Umbilicus: **T10** dermatome, at L3.
- **Scarpa's** (membranous) fuses with fascia lata ~1 cm below the inguinal ligament; continues as **Colles fascia** → dartos.
- Colles fuses to ischiopubic rami and perineal membrane → after **anterior urethral injury** blood/urine fill penis and scrotum = **"butterfly" perineal haematoma**; no spread to thigh, but free spread **upward** deep to Scarpa's to flank.
- †Buck's fascia intact → haematoma stays on the shaft; butterfly = Buck's breached.

Rectus sheath

	Above arcuate line	Below arcuate line
Anterior	External oblique + part of internal oblique	All three aponeuroses
Posterior	Rest of internal oblique + transversus	**Absent** — transversalis fascia + peritoneum only

- Rectus intersections tethered to anterior sheath → **transverse division without retraction**. Segmental nerves enter **laterally** → paramedian incision lateral to rectus → atrophy, ventral hernia. Pyramidalis absent in ~20%.

Inguinal canal

- ~**4 cm**; transmits **spermatic cord and ilioinguinal nerve**. Deep ring midway ASIS–pubic tubercle, **lateral to inferior epigastric vessels**; superficial ring superolateral to pubic tubercle.
- Walls: anterior external oblique; posterior transversalis + conjoint tendon; roof internal oblique/transversus; floor inguinal, lacunar ligaments, iliopubic tract. Roof contracts on floor = **shutter mechanism**.
- Direct hernia **medial**, indirect **lateral** to inferior epigastrics. **Radical orchidectomy**: ligate cord at internal ring → stump retracts into retroperitoneum.

Peritoneal folds (laparoscopic map)

Fold (medial → lateral)	Contains	Significance
Median	Urachus	Bladder apex → umbilicus

Fold (medial → lateral)	Contains	Significance
Medial (paired)	Obliterated umbilical arteries	Incise **lateral** to them → obturator fossa, retropubic space
Lateral (paired)	Inferior epigastric vessels	Deep ring; hernia type

3 · Pelvic floor and fascia

- Obturator internus fascia → **tendinous arch of levator ani** (ischial spine → pubis), origin of pelvic diaphragm (levator ani + coccygeus).
- Levator ani: **puborectalis** (anterior U-sling behind anorectal junction — **faecal continence**), pubococcygeus, iliococcygeus. Tonically contracted, type I + II fibres; relaxes to void. **Kegel exercises** augment tone.
- **Parietal fascia** lines walls; **visceral** sheaths organs; lateral attachments = **endopelvic fascia**, carrying pelvic vessels — gateway to the prostate.
- **Arcus tendineus fascia pelvis**: puboprostatic ligament → ischial spine; parietal–visceral junction; anchors urethral and bladder supports.

4 · Perineum

- Diamond; inter-tuberous line divides **urogenital** and **anal** triangles. **Perineal membrane**: triangle across pubic arch, pierced by urethra, supports sphincter.
- **Perineal body**: midline keystone for pelvic muscles/fascia, blends with Denonvilliers; injury at perineal prostatectomy → urinary + faecal incontinence.
- **Ischiorectal fossa** communicates with pelvis via sciatic foramina → perirectal infection spreads into pelvis.
- Perineal prostatectomy MCQ: **Young** divides central tendon, leaves external sphincter on rectum; **Belt** passes **under** the external sphincter.
- **Internal pudendal artery**: out greater, back via lesser sciatic foramen → **Alcock's canal** → dorsal penile artery; its veins join DVC via lateral prostatic plexus.
- **Pudendal nerve (S2–S4)**: external anal and striated urethral sphincters, levator ani, ischio-/bulbospongiosus; ends as dorsal nerve of penis.
- Penis: **stretched ≈ erect length**; shorter in ED.

5 · Pelvic vessels and lymphatics

- Common iliacs (anterior to veins) divide at the **SI joint**.
- **External iliac** → inferior epigastric and deep circumflex iliac. **Accessory obturator vessels** (from inferior epigastric) cross the obturator fossa — PLND hazard.
- **Internal iliac**: posterior trunk parietal (superior gluteal, ascending lumbar, lateral sacral); anterior trunk visceral (umbilical, obturator, middle rectal, inferior vesical, internal pudendal, inferior gluteal).
- Patent umbilical artery → **superior vesical** + artery of vas (collateral testicular supply). **Inferior vesical** (not internal pudendal) supplies the NVB. Obturator artery deep/medial to obturator nerve.
- External and internal iliac veins join **behind the internal iliac artery** — major PLND bleed.

- **DVC** enters under symphysis anterior to the striated sphincter; ligation at radical prostatectomy can damage the sphincter laterally. **Santorini's plexus** under lateral prostatic fascia.
- Lymph: frequent **right–left cross-drainage**. External iliac — bladder, urachus, wall; internal iliac — prostate, bladder. **Node of Cloquet = distal limit of PLND**.

6 · Pelvic nerves

Nerve	Roots	Key point
Ilioinguinal	L1	Through inguinal canal → anterior scrotum
Genitofemoral	L1–L2	On psoas — **lateral limit of PLND**; genital branch → cremaster
Femoral	L2–L4	Psoas hitch, retractor blades, inguinal dissection
Obturator	L2–L4	Iliac bifurcation → obturator canal; PLND injury → **weak adduction**
Pudendal	S2–S4	Perineal sensation; levator, urinary and anal sphincters
Nervi erigentes	S2–S4	Parasympathetic to bladder, rectum

> **KEY POINT**
> Anterior scrotum is L1 (ilioinguinal, genitofemoral); posterior scrotum is sacral (pudendal, posterior femoral cutaneous).

- P1 Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer — *pelvic plexus (T10–L2 + S2–S4) beside rectum at seminal vesicles → cavernous nerves, NVB outside Denonvilliers at 5 and 7 o'clock*
- NVB at **5 and 7 o'clock** at the apex; injury at prostatectomy or rectal resection → ED.

7 · Pelvic viscera

Rectum

- Begins at **S3**. Incising the **rectovesical pouch** exposes seminal vesicles (posterior robotic approach); **Denonvilliers fascia** separates rectum from prostate.
- Distal rectum near apex = **commonest rectal injury site** at prostatectomy.

Bladder

- Extraperitoneal except dome → **urachus** (adenocarcinoma, fistula, cyst). †Space of Retzius = extraperitoneal access. Infant bladder intra-abdominal.
- Detrusor: inner longitudinal, middle circular, outer longitudinal. Bladder-neck circular fibres = **internal sphincter**, **α1-adrenergic** → continence and **antegrade ejaculation**; damage adds to post-prostatectomy incontinence.
- Body: parasympathetic cholinergic; NANC fibres → *Ch 103*.
- Pedicles posterolateral; veins → internal iliac. Lymph mainly **external iliac**, also internal iliac, obturator, bilaterally.

Ureterovesical junction

- Intramural ureter in **Waldeyer's sheath** **1.5–2 cm** oblique; narrow → stone impaction.
- **Flap valve**: detrusor backing compresses intramural ureter. Weak backing → **VUR**; weak hiatus → **Hutch diverticulum**.
- Superficial trigone anchors at verumontanum; deep trigone (Waldeyer's + detrusor) at bladder neck.

- Pelvic ureter: narrows at iliac crossing, under umbilical artery, **behind the vas**. Blood supply enters **laterally** → keep lateral attachments; stripping → stricture.

8 · Endoscopic anatomy

ALGORITHM

Flexible cystoscopy (local) → bladder, orifices, trigone → retroflex at neck (intravesical prostatic protrusion, hidden tumours) → verumontanum with ejaculatory ducts → external sphincter coapts just distal → bulbar urethra

KEY POINT

Verumontanum = sentinel at TURP: resection distal to it injures the external sphincter → incontinence.

9 · Radiographic anatomy

- **Phlebolith**: venous calcification mimicking distal ureteric/bladder stone.
- **Cystogram**: absent lateral distension → extraperitoneal pelvic haematoma or post-prostatectomy lymphocele.
- **VCUG** best shows posterior urethra (sphincters relaxed). Oblique **RUG**: steady pressure overcomes external sphincter.
- Membranous urethra = **shortest, narrowest** segment — not a stricture. Opacified Littré/Cowper's ducts normal.
- †Pelvic fracture urethral injury: bulbomembranous junction.
- **CT**: high spatial, low contrast resolution → contrast needed except for **stones**. **Extraperitoneal rupture** urine in retropubic/perivesical spaces.
- Malignant node: rounded, heterogeneous; size most specific at ≥ **1 cm** short axis.
- **CT urogram**: excretory images at **7–10 min**; iliac-crossing narrowing, peristaltic gaps normal.
- **MRI**: best soft-tissue contrast; T1 bone/gadolinium, T2 anatomy, prostate, pelvic floor. Distended bladder wall > **3 mm** = thickening. External sphincter = T2-dark doughnut below apex.

103 Physiology and Pharmacology of the Bladder and Urethra

1 · Overview

- Lower urinary tract (LUT) has two functions: **storage** (continence) and **emptying** (micturition), integrating CNS, peripheral nerves, detrusor smooth muscle (DSM), urothelium, urethra, pelvic floor and external urethral sphincter (EUS).

> **KEY POINT**
> **Why single-organ therapy fails:** bladder and urethra act reciprocally — boosting contractility in detrusor underactivity fails unless the urethra relaxes. Afferent modulation acts on both (reduce for overactivity, augment for underactivity) → most promising therapeutic lever.

2 · Functional anatomy

Bladder

- **Body** above ureteric orifices; **base** = trigone + bladder neck — differ in neuropharmacology.
- **Male bladder neck**: complete smooth muscle ring, dense sympathetic supply; closes in ejaculation → **antegrade ejaculation**. Female: no collar, sparse adrenergic supply.
- **Lamina propria** = "functional centre": afferent plexus (TRPV1, P2X3), interstitial myofibroblasts, subepithelial capillaries.
- **Blood flow** falls with distension (wall tension); more so in low compliance. Ischaemia–reperfusion → patchy denervation, as in detrusor overactivity (DO).

Stroma

- Wall ≈ **50% collagen, 2% elastin**; types I, III, IV.
- **Poorly compliant bladders**: ↑ connective tissue : muscle ratio, ↑ **type III : type I collagen** → ↓ capacity.

Urethra and sphincters

- Urethra has smooth muscle (lissosphincter) and an **intramural striated rhabdosphincter (EUS)**, separate from periurethral pelvic floor.
- Male EUS: crescent ventral to prostate → horseshoe distally. Female EUS: horseshoe over ventral urethra; subepithelial vascular plexus = **"urethral seal"**.
- Levator ani does not encircle the ventral urethra — direct continence role questioned.

Feature	Slow-twitch	Fast-twitch
Role	Sustained resting tone	Rapid extra squeeze on coughing
Fatigue / metabolism	Slow; oxidative	Rapid; anaerobic
Where	Intramural rhabdosphincter — **~50% of static urethral resistance**	Periurethral pelvic floor (mixed)
Rhabdosphincter mix	Male **65% slow / 35% fast**; female **87% slow / 13% fast**	

3 · Urothelium: barrier and sensor

Barrier

- ~7 layers; apical **umbrella cells**. **Uroplakins** form **asymmetrical unit membrane** plaques — the barrier, and binding site of **type 1-fimbriated *E. coli***.
- **Tight junctions** (occludin, claudins, ZO-1) block paracellular flow; stretch lowers ionic resistance tenfold without altering small-molecule permeability.
- Umbrella cell loss (infective/radiation cystitis) → solute leak → symptoms worse with distension; hyperosmolality excites C fibres.
- **GAG layer**: "antiadherence factor" = mucin MUC-1; chondroitin sulfate main human GAG; **not proven** as primary barrier.

Sensor–transducer function

Urothelial cells act like sensory neurons (receptors for bradykinin, TrkA/p75, P2X/P2Y, α/β, nicotinic/muscarinic, ENaC, TRPV1/2/4, TRPM8); on stretch they release mediators onto afferents, directly or via myofibroblasts.

Mediator	Released by / when	Effect
ATP	Stretch; ↑ injury, inflammation, ageing	Excites suburothelial **P2X3** afferents → fullness, pain; P2X3-null mice hyporeflexic
NO	Especially inflammation	Inhibits afferent Ca^{2+} channels → **afferent inhibition**
ACh (non-neuronal)	Basal; ↑ stretch, ageing	Modulates afferents/DSM → antimuscarinics may act partly here
Prostaglandins	Urothelium main source	Detrusor regulation, cytoprotection
Inhibitory factor (unknown)	Basal	↓ muscarinic contraction; not NO or PG

- Afferent threshold = **ATP (excitatory) : NO (inhibitory) balance**; ATP ↑ in DO and BPS/IC.
- **Suburothelial myofibroblasts**: vimentin+, α-SMA+, desmin−; **Cx43** gap junctions; respond to ATP (P2Y6), ACh (M2/M3); M2/M3 expression correlates with urgency in idiopathic DO.

> **KEY POINT**
> Damaged urothelium causes symptoms twice over — **solute leak** into lamina propria and **altered signalling** (more ATP, less NO) to afferents.

4 · Detrusor smooth muscle physiology

- Randomly oriented fascicles; no sarcomeres — actin/myosin anchor to **dense bodies**. Not every cell innervated — **gap junctions** spread excitation.
- Vs skeletal: autonomic + paracrine control; Ca^{2+}–calmodulin → **myosin phosphorylation** (slow, "latch state"); mostly tonic.

> **ALGORITHM**
> ACh → **M3** (Gq) → ↑ $[Ca^{2+}]i$ (L-type Ca^{2+} entry + PLC → IP3 → SR release) → Ca^{2+}–calmodulin → MLCK → phosphorylates MLC20 → cross-bridge cycling → contraction → MLCP dephosphorylates as Ca^{2+} falls → relaxation

- Tone = Ca^{2+} + **MLCK : MLCP balance**. **Calcium sensitisation**: RhoA → **Rho kinase** and PKC (CPI-17) inhibit MLCP → force without ↑ Ca^{2+}.

- **L-type VDCC** = main contraction trigger (nifedipine, diltiazem, verapamil); T-type fine control.
- Resting −50 to −60 mV; upstroke Ca^{2+}, repolarisation K^+ (SK, **BK**, Kv). **BK is the key brake** (Ca^{2+} sparks → BK → hyperpolarisation); BK knockout → OAB; BK ↓ in NDO.

G protein	Receptors	Effect
Gq	M1, M3, M5 ("odd"); α1; AT II; H1; 5-HT2	IP3 + DAG → Ca^{2+}, PKC → contraction
Gi/o	M2, M4 ("even")	↓ cAMP
Gs	β3	↑ cAMP → PKA → relaxation

- **Purinergic (P2X1) contraction**: minimal in normal human bladder; appears in obstructed, OAB, ageing bladders. It suppresses muscarinic coupling → **DHIC**: urgency + incomplete emptying.
- Detrusor deliberately **poorly coupled** (no tetanic contractions) → protects filling. ↑ Cx43 (BOO, NDO; cytokines, NGF) → whole-bladder myogenic contraction = DO.
- **Interstitial cells**: c-KIT+ (imatinib ↓ spontaneous contractions; ↑ OAB/BOO) and inhibitory PDGFRα+ (P2Y1).

KEY POINT

Motor–sensory coupling: spontaneous micromotions generate afferent bursts perceived as urgency. Antimuscarinics and mirabegron reduce non-voiding contractions and so sensory input — efferent and afferent functions are inseparable.

5 · Receptors and drug targets

Receptor	Location	Function	Drug target
M3 (Gq)	Detrusor; urothelium; salivary glands	**Mediates contraction**	Antimuscarinics
M2 (Gi)	Detrusor (**numerically predominant**)	Amplifies M3 (opposes β-cAMP); shift M3 → M2 in neurogenic/obstructed bladder	Non-selective antimuscarinics
β3 (Gs)	Detrusor (**most expressed adrenoceptor**); afferents	Storage relaxation; ↑ compliance	Mirabegron, †vibegron
α1A	Urethra, prostate	Urethral closure; female resting tone	α-blockers (BPH); α-agonists ↑ closure
α1D	Bladder (predominant α); spinal cord	Storage symptoms of obstruction	Tamsulosin improves storage LUTS
α2	Prejunctional urethra; ganglia	↓ noradrenaline release	—
P2X3	Suburothelial afferents	Senses ATP; ↑ in BPS/IC	AF-219 (proof-of-concept)
P2X1	Detrusor	Purinergic contraction; ↑ BOO, ageing	Potential for DHIC
TRPV1	C fibres; urothelium	Heat >43 °C, protons, capsaicin; ↑ in NDO	Capsaicin, RTX desensitise
TRPM8	Afferents	Cold → bladder cooling reflex	Experimental
NK1	Spinal cord	Afferent transmission	Aprepitant ↓ urgency (trial)

Receptor	Location	Function	Drug target
Nicotinic	EUS	Sphincter contraction	—

- P3 OAB Pharmacotherapy: Darifenacin vs Solifenacin vs Tolterodine, and Mirabegron — *tissue > subtype selectivity; darifenacin ~30-fold M3; β3 → cAMP; dementia and antimuscarinic burden*
- **Oxybutynin**: metabolite **desethyloxybutynin** causes most dry mouth → controlled-release form better tolerated.
- CNS-penetrant agents block M1 → delirium, cognitive decline; reverses within weeks of stopping.
- Other storage drugs: Ca^{2+} antagonists, K^+ openers, imipramine (anticholinergic + direct relaxant + noradrenaline reuptake block), capsaicin/RTX. Outlet: **α-agonists** (ephedrine, midodrine); **oestrogen** ↑ seal and blood flow but disappointing alone; systemic oestrogen may worsen UI.
- P3 Duloxetin — *SNRI ↑ Onuf's nucleus output → ↑ sphincter tone; mild–moderate SUI; Europe, not USA*

6 · Peripheral innervation

- P1 Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children — *pelvic (S2–S4) voiding, hypogastric (T10–L2) storage, pudendal/Onuf's EUS; Aδ vs silent C fibres; storage spinal vs spinobulbospinal voiding reflex; PAG–PMC; central transmitters*
- Parasympathetic **S2–S4**; sympathetic †**T10–L2**. Sympathetic ganglia: chain → inferior mesenteric → pelvic.
- Parasympathetic postganglionic neurons partly **intramural** → cauda equina/pelvic plexus injury **decentralises rather than denervates**.
- Afferent plexus densest at bladder neck, absent at dome. Filling → episodic bursts; last perceived as urgency.
- Aδ: muscle tension receptors, capsaicin-resistant. C: mucosal, mostly **silent**, capsaicin/RTX-sensitive; drive new **spinal reflex** after SCI, inflammation, BOO, ageing.
- **Sensitisation** by ATP, NGF (TrkA), PGs, bradykinin, low pH, high osmolality → central sensitisation. **Cross-organ sensitisation** explains BPS/IC–IBS overlap.
- μ-opioids ↓ sensation, ↑ sphincter tone → postoperative retention.

7 · Central control

- **PMC = Barrington's nucleus** = "M region" (dorsal pontine tegmentum). Lesions (stroke, MS, tumour) → **retention**. PMC does **not** project to Onuf's — relaxes sphincter **indirectly** via GABA/glycine interneurons.
- **L-region** (ventrolateral pons) = storage centre → Onuf's nucleus.
- **PAG** relays afferents to PMC: binary "on–off" switch; without higher control bladder empties at a critical volume (infant).
- **Forebrain**: PAG → thalamus → **insula** (desire) → **prefrontal cortex** (decision) → **tonic suppression** of reflex. Frontal white-matter lesions → **incontinence**; anterior cingulate and SMA over-respond in urge incontinence.

> **KEY POINT**
> **Lesion localisation:** forebrain → loss of tonic inhibition → DO, coordination preserved; pontine → retention; suprasacral cord → DO + **DSD**; sacral/cauda → decentralised, acontractile bladder → *Ch 104*.

8 · Storage and emptying phases

- Low storage pressure: intrinsic wall properties, quiet parasympathetic outflow, and a **sympathetic storage reflex** — weak in humans (bilateral RPLND does not alter storage).
- **Guarding reflex**: filling recruits pudendal motoneurons (EUS EMG rises). On cough, urethral pressure rises **before** pressure transmission → active neural component. Suppression in voiding is supraspinal → lost after SCI → **DSD**.
- **Sphincter-to-bladder reflex**: pudendal afferents inhibit bladder preganglionics — basis of "holding on" and neuromodulation.

> **ALGORITHM**
> Tension-receptor (Aδ) firing exceeds threshold → PAG/PMC switch flips (if cortex permits) → sympathetic and pudendal outflow off → **urethral sphincter relaxes first** + urethral smooth muscle relaxes (NO) → detrusor contracts (parasympathetic) → flow → urethral afferents reinforce contraction until empty

- **Urethra-to-bladder reflexes** (Barrington): flow facilitates contraction; links SUI (urine in proximal urethra) with detrusor instability.
- After SCI: spinal shock (weeks of retention) → NGF-driven **C-fibre spinal reflex** → NDO with DSD.
- **Pdet = Pves − Pabd**. Power ≈ Pdet × Q roughly constant → voiding **Pdet reflects outlet resistance, not contractility**.
- SUI women void at Pdet at Qmax ≈ **19 cmH₂O**; healthy men ≈ **60–70 cmH₂O**. Low pressure in women ≠ impaired contractility; male nomograms not valid in women.

9 · Urethral closure mechanisms

- Nicotinic blockade rarely lowers tone >40%; α-blockade lowers urethral pressure by about **one third** → smooth muscle (α1A) tone matters.
- EUS closes uniformly like an **iris**; noradrenaline augments it.
- **Mucosal seal**: vascular filling of lamina propria; oestrogen ↑ blood flow; poor arterial supply ↓ urethral pressure.
- **Passive pressure transmission** to proximal urethra; **hammock hypothesis (DeLancey)**: urethra compressed against a firm posterior fascial layer.
- **Active** guarding: fast-twitch reflex squeeze on coughing.

> **KEY POINT**
> Continence = **active muscle tone + passive anatomical coaptation**. Integral theory → mid-urethral sling (MUS), the SUI standard, yet DeLancey's data suggest intrinsic sphincter function matters more than support *→ Ch 109*.

10 · Compliance

- P3 Urinary bladder compliance — *C = ΔV/ΔP; collagen/elastin, coiled collagen straightens; fast filling falsely low; DLPP >40 cmH₂O → upper tract damage*
- Physiological filling keeps pressure **<10 cmH₂O**. Determinants: passive viscoelasticity, active tone, geometry; **type III collagen straightens** with filling.
- Pelvic nerve (not acute spinal) transection reduces compliance → afferent–CNS loop inhibits contractility during filling.

- †DLPP ≥ **40 cmH₂O** in low-compliance neurogenic bladder predicts upper tract damage *→ Ch 104*.

11 · Pathophysiology links

- **Idiopathic OAB**: DO in a minority (11% in one cohort) → dysregulated sensation ("afferent neurourology"); ↑ ATP/TRPV1, ↑ Cx43, ↓ BK, NGF plasticity.
- **BOO**: hypertrophy, partial denervation → **muscarinic supersensitivity**, ↑ P2X1, Cx43 (myogenic theory, Brading). NGF stays ↑ → **persistent storage symptoms after prostate surgery**.
- **BPS/IC**: ↑ NGF, ATP/P2X3, TRPV1, mast cells; tanezumab and intravesical RTX failed (heterogeneity) *→ Ch 113*.
- **UAB** = symptom complex; detrusor underactivity = urodynamic finding; afferent loss may contribute. PVR ≥ **300 mL** warrants evaluation (also bothersome LUTS, hydronephrosis, renal insufficiency, culture-proven UTI); PVR <300 mL does not prove normal contractility.

12 · Ageing

- **Hypofunction**: ↓ contractility, sensation, urethral closure pressure. Capacity not reduced per se — small capacity = coexisting DO.
- ↓ muscle : collagen; ischaemia; ↓ cholinergic nerves. **α-contraction ↑ (α1D), β-relaxation ↓** → noradrenaline contracts the aged bladder.
- Aged CNS prone to overactivity → urgency with poor emptying (DHIC). †Prefer β3 agonists or low-CNS-penetrant agents in elderly.

13 · Neuromodulation and botulinum toxin — mechanisms

- **SNM**, pudendal and PTNS treat refractory OAB **and** non-obstructive retention; target = **afferent axons**. OAB: somatic input inhibits preganglionics/PMC transfer; retention: switches off overactive **guarding reflex**.
- Pudendal stimulation carries S2–S4 afferents, avoids S3 sciatic discomfort.
- **OnabotulinumtoxinA**: heavy chain binds terminal → light chain (zinc protease) cleaves **SNAP-25** → ↓ ACh release; also ↓ TRPV1/P2X3, SP, CGRP, urothelial ATP.
- RTX desensitises TRPV1 **without prior excitation**, unlike capsaicin.

14 · Landmark trials

Trial	Design	Result	Take-home
ABC	RCT: intradetrusor OBTX vs oral antimuscarinic, idiopathic OAB with UUI	Equal ↓ in daily UUI; OBTX more complete dryness	Antimuscarinic: dry mouth; OBTX: catheterisation, UTI
ROSETTA	RCT: OBTX 200 U vs SNM, women with refractory non-neurogenic OAB, 2 years	Comparable efficacy; OBTX more often dry	OBTX more UTIs (note 200 U dose)

104 Pathophysiology and Classification of Lower Urinary Tract Dysfunction

1 · Normal function: two phases

Storage requires	Voiding requires
Low detrusor pressure (normal compliance), appropriate sensation	Detrusor contraction of adequate magnitude and duration
Outlet closed at rest and on straining	Fall in smooth and striated sphincter resistance
No detrusor overactivity (DO)	No anatomic obstruction

- **Storage mostly passive** (viscoelastic stroma). ↑ collagen (inflammation, BOO, decentralisation) → low compliance, **drug-unresponsive**.
- Storage reflexes: α1 bladder neck tone, β3 detrusor relaxation, spinal **guarding reflex**. Voiding: pelvic parasympathetic, coordinated by the **pons**.
- **Stress continence**: equal pressure transmission to proximal urethra (**Enhorning**) + active reflex closure (**Tanagho**); plus **mucosal seal** (poor if scarred/atrophic). **Klevmark**: cystometric pressure rise is a fast-filling artefact.

2 · Pathophysiology

Every failure is **bladder, outlet, or both** — basis of the **Wein functional classification**.

Storage failure

- **Bladder overactivity** = involuntary contractions and/or low compliance (neurological, BOO, ageing, idiopathic).
- **OAB** (ICS): urgency ± UUI, usually with frequency, nocturia. UUI → assume DO (UDS often miss it).
- **Hypersensitivity** without overactivity: IC/BPS. **Outlet underactivity**: denervation, surgery/trauma, ageing, loss of female support.

	Hypermobility (genuine SUI)	Intrinsic sphincter deficiency (ISD)
At rest	Outlet competent	Bladder neck poorly functional/open
Cause	Poor pelvic support	Prior surgery, scarring, neurological
Treatment	Correct hypermobility	Coaptation/compression

> **KEY POINT**
> UI can occur from ISD alone, **never from hypermobility alone**. **DeLancey's hammock**: firm suburethral layer lets abdominal pressure compress the urethra. In half of continent women the continence point is the **mid-urethra**. Male sphincteric incontinence resembles ISD.

Emptying failure

- **Bladder underactivity**: neurogenic, myogenic (overdistension, drugs), psychogenic; diabetes = neuromuscular + vascular.
- **BOO** (ICS): low flow and/or raised PVR **with high detrusor pressure**. Male: prostate, bladder neck contracture, stricture; female: prolapse, post-sling.

- **Functional**: striated sphincter dyssynergia (commonest); smooth sphincter dyssynergia (**autonomic dysreflexia**); primary bladder neck obstruction (young men); dysfunctional voiding (**non-neurogenic**).

3 · Functional classification (Wein) — 2 × 2 table

	Because of the bladder	Because of the outlet
Failure to store	DO; low compliance; hypersensitivity; underactivity with overflow	Hypermobility; ISD. (Plus **fistula**)
Treatment	Bladder training; antimuscarinics, β3-agonists, onabotulinumtoxinA; neuromodulation; augmentation	Pelvic floor training; α-agonists, duloxetine; bulking; slings; AUS
Failure to empty	Underactivity: neurogenic, myogenic, psychogenic	Anatomic obstruction; sphincter dyssynergia; bladder neck dysfunction
Treatment	Valsalva/Credé; parasympathomimetics; electrical stimulation	α-blockers, 5ARIs; prostatectomy, incision; sphincter onabotulinumtoxinA, sphincterotomy
Circumvent	Pads/collecting devices, antidiuretics, catheterisation, diversion	

KEY POINT

- Bladder underactivity appears under **both** storage (overflow) and emptying. MCQ
- α-agonist tightens outlet (storage); α-blocker relaxes it (emptying).
- OnabotulinumtoxinA: intradetrusor → storage; sphincteric → emptying.
- Naming the failure names the treatment family.

Combined deficits (T10 paraplegia: DO + dyssynergia): external collection, or **abolish DO** + CIC.

4 · Urodynamics by phase and site

Phase	Bladder	Outlet
Storage	Filling cystometry; DLPP	Urethral pressure profile; VLPP
Voiding	Voiding cystometry (Pdet)	Fluoroscopy; sphincter EMG
Both	Flowmetry and PVR — **cannot localise** the fault	

5 · Other classifications

- P2 Pathophysiology, classification and management of LUTS. — *ICS storage / voiding (incl. terminal dribble) / post-micturition (post-micturition dribble) symptoms; Wein summary*
- **Signs (ICS)**: bladder diary, examination (prolapse may mask UI), pad test.

ICS urodynamic classification (most precise)

Domain	Storage	Voiding
Detrusor	Normal; DO (neurogenic/idiopathic)	Normal; underactive; acontractile
Sensation, capacity, compliance	Normal, high/increased, low/reduced, absent	—
Urethra	Normal or incompetent closure	Normal; BOO, dysfunctional voiding, DSD

- Cystometric capacity = permission to void; **maximum cystometric capacity = can no longer delay**; maximum anaesthetic capacity = under anaesthesia.
- DSD = detrusor contraction with involuntary striated sphincter contraction; "neurogenic" prefix when a relevant neurological disorder exists.

† **Lapides**: sensory neurogenic, motor paralytic, uninhibited, reflex, autonomous. **Bors–Comarr**: upper/lower motor neuron, complete/incomplete, balanced/unbalanced. MCQ

6 · Nocturia and neurogenic dysfunction

- **Polyuria** > **40 mL/kg/24 h**. **Nocturnal polyuria**: night ÷ 24-h volume > **20% young, >33% elderly**; loss of nocturnal vasopressin rise or sleep apnoea.
- **IC/BPS**: normally silent **C fibres** hyperactive (central sensitisation); Aδ signal distension. Bladder–bowel crosstalk → OAB–IBS overlap.

Lesion	Detrusor	Sphincter
Suprapontine	DO	Synergic
Suprasacral spinal	DO	DSD, raised PVR
Sacral	Acontractile	Impaired

KEY POINT
DSD needs an intact sacral arc **disconnected from the pons** — hence suprasacral, never suprapontine.

106 Evaluation and Management of Males With Urinary Incontinence

1 · Definition and determinants

- **UI** (ICS): any involuntary loss of urine; a storage LUTS *→ Ch 104*. Male continence: **EUS most important**.
- Risk factors: **age (major, independent)**, LUTS, infection, cognitive/neurological impairment, obesity, diabetes, prostatectomy, drugs, smoking.
- Reversible causes — **DIAPPERS**: Delirium, Infection, Atrophic urethritis, Pharmacological, Psychiatric, Endocrine, Reduced mobility, Stool impaction.
- Male:female ≈ 1:2; **UUI predominates** > SUI > MUI. Often multifactorial → establish aetiology first.

2 · Types

Type (ICS)	Mechanism	Male context
SUI — effort, cough	Sphincter deficiency	Almost only **after prostatectomy**
UUI — with urgency	DO overcomes normal outlet	~**90%** have DO on urodynamics; = wet OAB
MUI	DO + sphincter deficiency	Uncommon; may follow prostatectomy
Nocturnal enuresis	Wetting in sleep	New in adult man → suspect **high-pressure chronic retention**, renal failure
Continuous	Fistula, exstrophy/epispadias, ectopic ureter	Rare
With chronic retention	Residual ≥**300 mL**	"Overflow" no longer ICS term; SCI ≥T6 → autonomic dysreflexia risk

- Others: postural, insensible, **postmicturition dribble**, functional, situational.

3 · Evaluation

- History: timing, precipitants, containment, prior prostate/pelvic surgery, neurology, bowel/sexual function, drugs, **most bothersome symptom**. Examine abdomen, genitalia (meatal stenosis, phimosis), DRE, anal tone, neurology, cognition; **cough test supine then standing**.
- **BMI**: weight loss may help SUI. With an AUS, abdominal pressure above balloon pressure (61–70 cm H_2O) leaks through the cuff.
- **Dipstick** in all; haematuria → cytology, cystoscopy, upper-tract imaging.
- **Bladder diary**: only fully validated = ICIQ-BD; **3 days = 7 days**, include a weekend.
- IPSS does **not measure incontinence** → **ICIQ-UI Short Form** (0–21). **Pad test** not routine; cannot separate UUI from SUI.

Severity	Pads/day	Pad weight/day
Mild	1	<200 g
Moderate	2–4	200–400 g

Severity	Pads/day	Pad weight/day
Severe	>4	>400 g

- Pad weight beats count; neither validated.
- **Flow + PVR**: fast rise → OAB; slow rise, long tail → BOO; flow cannot separate BOO from underactivity. One representative flow acceptable (ICS 2017).
- PSA only if life expectancy >10 years and it would change management.
- Imaging/cystoscopy: high PVR, haematuria, stricture. **UDS only if it alters management** — before invasive treatment (ICI, EAU, AUA) → *Ch 107*.
- SUI = symptom; **USI** = leak on effort without detrusor contraction; **cough-associated DOI** mimics SUI → fill supine and cough: leak without DO = USI.

KEY POINT
Every man: history, examination, BMI, dipstick, diary, questionnaire, flow, PVR. Optional: PSA, bloods, imaging, endoscopy, invasive UDS — usually just before surgery.

4 · Urgency urinary incontinence

ALGORITHM
Lifestyle + bladder training + PFMT ≥6 weeks (ideally 3 months) → antimuscarinic or β3-agonist (± combination) → UDS optional → onabotulinumtoxinA / SNM / PTNS → augmentation or ileal conduit

- Cut fluid by **25%** (if intake >1 L/day); avoid caffeine, fizzy drinks; weight loss.
- **Antimuscarinics**: similar efficacy; ER preferred. **Anticholinergic burden** ≥3 months → dementia risk ~50% ↑, falls. Trospium — little brain penetration → *Ch 113*.
- **β3-agonists** co-first-line, less dry mouth. Mirabegron — **monitor BP**; vibegron — no CYP450 interaction. **Cognitive impairment → β3-agonist or early surgery**.
- Mirabegron + solifenacin beats escalating solifenacin. α-blocker + antimuscarinic/β3 for mixed LUTS. Tadalafil: only PDE5i for LUTS with ED.

	OnabotulinumtoxinA	SNM	PTNS
Delivery	Cystoscopic, **100 U** (200 U neurogenic DO)	S3 lead, implanted generator	Needle behind medial malleolus, 30 min weekly × 12
Strength	Repeat 6–9 months	Most effective, durable >10 years	Least invasive; better for dry OAB
Drawback	Retention ~**10%** → must accept self-catheterisation	Most invasive, costliest	Frequent visits

- SNM test: ≥**50%** improvement → permanent generator → *Ch 115*. No head-to-head trials.
- Refractory: containment or augmentation/ileal conduit (high morbidity).

5 · Stress urinary incontinence

ALGORITHM
Supervised PFMT ≥3 months (± biofeedback, electrical stimulation; better started pre-op) → ± duloxetine → exclude anastomotic/urethral stricture and bladder dysfunction (UDS) → surgery by severity

- P2 Post-prostatectomy incontinence — *wait 6–12 months + PFMT; duloxetine off-label; no bulking; sling mild–moderate, AUS (cuff, balloon, scrotal pump) severe/irradiated; ProACT; MASTER*
- After radical prostatectomy wait **6–12 months**; avoid long-term urethral catheter (traumatic hypospadias).
- **AUS = gold standard**, ~80% continence (Scott 1973 MCQ); activate ~6 weeks; †deactivate before catheterisation/endoscopy (erosion).
- Slings: fixed transobturator (AdVance) or adjustable (Remeex, Argus-T, ATOMS) — neither superior. Radiotherapy, severe leak, prior TUR worsen fixed-sling results → AUS.
- **ProACT**: bladder-neck balloons; explant up to half; **contraindicated after radiotherapy**.
- Neurogenic SUI often needs several procedures → *Ch 109*.

EAU 2023 recommendation (PPI)	Grade
Offer AUS for moderate–severe PPI	Strong
Warn of AUS/ProACT complications, mechanical failure, explantation	Strong
Inform of duloxetine side effects, off-label use	Strong
Fixed transobturator sling for mild–moderate PPI	Weak
Do not offer bulking agents; no ProACT after radiotherapy	Weak

6 · Mixed and other types

- **MUI**: treat most bothersome component first; DO may mask SUI → treat DO, repeat UDS.
- High-pressure chronic retention with enuresis: **catheterise first**, then relieve BOO.
- Enuresis with low residual → OAB → antimuscarinic/β3 ± desmopressin.
- Postmicturition dribble: post-void PFMT squeeze + **urethral milking** (press behind scrotum forward to penile base).
- **Red flags** → specialist: haematuria, pain, recurrent infection, pelvic radiotherapy, abnormal DRE, high PVR. AUS: ICI grade A.

7 · Landmark trials

Trial	Design	Result	Take-home
MASTER	Non-inferiority RCT, male sling vs AUS for urodynamic SUI after prostatectomy	Equivalent at 12 months; secondary outcomes favoured AUS; few completely dry	Sling acceptable alternative; counsel realistically

107 Urodynamic and Videourodynamic Evaluation of the Lower Urinary Tract

1 · Definition and role

- **Urodynamics (UDS)**: dynamic study of transport, storage and evacuation of urine. ICS separates UDS *studies* (tests) from *observations*.
- Use when history, examination and simple tests are insufficient: (1) **diagnose the cause of symptoms**; (2) assess **diseases that silently damage upper tracts** (SCI, MS, radiation cystitis). Also predicts outcome, explains treatment failure.

Wein functional classification — the frame for every study

Site	Overactive	Underactive
Bladder	Failure to store	Failure to empty
Outlet	Failure to empty	Failure to store

- Set the questions first, then design the study. Aim to **reproduce the symptom**; an abnormality without symptom reproduction is not necessarily diagnostic, and failure to record one does not exclude it.

> KEY POINT
> **Box 107.1 — Urodynamic risk factors** (treat even if asymptomatic): impaired compliance; DESD; DISD; high-pressure DO throughout filling; DLPP **> 40 cmH_2O**; poor emptying with high storage pressure — each a route to sustained high intravesical pressure.

2 · Preparation

- **Medication status** decided in advance: assess drug effect → test on regular dosing; uncover aetiology → stop drug first.
- **ICS pre-invasive work-up**: symptom/bother score, **3-day frequency-volume chart**, free flow and PVR (normal free flow lets an abnormal invasive flow be recognised as artefact).
- **Urinalysis before every invasive study**; symptomatic UTI → postpone. Commonest after-effect: dysuria.

Antibiotic prophylaxis (AUA / SUFU Best Practice Policy)

"Index" patient: **no routine prophylaxis** (level 1). Non-index: **single dose within 1 h** — first choice oral trimethoprim–sulfamethoxazole; alternatives cephalosporin, amoxicillin–clavulanate, aminoglycoside ± ampicillin, fluoroquinolone.

Prophylaxis YES	Prophylaxis NO
NLUTD; BOO or raised PVR; age > 70; **asymptomatic bacteriuria**; immunosuppression (esp. renal transplant); chronic catheters incl. CIC; high-risk orthopaedic implant	Diabetes; **recurrent UTI history**; recent hospitalisation; obesity; menopause; gender; prior urological surgery; cardiac valve disease (not routinely)

Trap: asymptomatic bacteriuria is treated here, unlike most urological settings.

3 · Non-invasive components

- Bladder diary (3 days): functional capacity, frequency, nocturnal output → sets cystometric target and checks a flow is representative.
- P2 Uroflowmetry. — *Qmax, Qave, voiding efficiency; VV ≥150 mL; Siroky/Liverpool nomograms, corrected Qmax; bell, plateau (stricture), intermittent curves; cannot separate BOO from underactivity*
- Minimum **VV 150 mL** (Qmax linear above, hyperbolic below). †Qmax > 15 mL/s men (< 10 suggests obstruction), > 20 mL/s women.
- **PVR**: scan or catheter (state method); repeat if raised; no agreed threshold.

4 · Equipment, transducers and quality control

- Minimum: Pves, Pabd, subtracted Pdet, flow; optional EMG, urethral pressure. Infusate saline or contrast; **not CO_2**.

Feature	External water-filled	Air-charged	Microtip
Reference level	**Upper border of symphysis pubis**; re-level on position change	None (no hydrostatic effect)	—
Problems	Air bubbles; motion, hydrostatic artefact; underdamped	Overdamped (> 3 Hz attenuated) — blunts coughs	Expensive, reusable
Standards	Basis of most nomograms	No reference values	—

- Systems **not interchangeable** — Pdet may differ by up to 10 cmH_2O; patterns transfer, absolute values do not.

KEY POINT

Quality control † (ICS Good Urodynamic Practice): zero to atmosphere; flush air; resting Pves and Pabd physiological and similar (Pdet near 0); **cough test** before filling, at intervals and before voiding — equal spikes confirm transmission (biphasic Pdet spike → re-flush/reposition). Artefacts: rectal contractions (unexplained fall in Pdet), catheter displacement, kinked tubing, stream hitting funnel edge.

5 · Filling cystometry (CMG)

- Filling to "permission to void" = **maximum cystometric capacity (MCC)**. **Pdet = Pves − Pabd**; Pabd by rectal catheter preferred.
- Rates: standard adult **30–100 mL/min**; †ICS physiological < body weight (kg)/4 mL/min. Neurogenic: physiological rate, body-warm saline (EAU). DO commoner upright.
- **Normal ("stable")**: Pdet near zero, no involuntary contractions. **Sensation** = only subjective parameter: †first sensation, first desire, strong desire, urgency.

Detrusor overactivity (DO)

- Involuntary detrusor contractions (IDC) during filling. **Idiopathic** vs **neurogenic** (relevant neurological disorder) — label from history, tracings identical. "Hyperreflexia"/"instability" obsolete.
- Absence doesn't exclude: **up to 50%** of women with UUI show no DO. Seen in some asymptomatic volunteers.
- Confirm involuntary; record volume, pressure, leak; characterise phasic/terminal, suppressible.

- BOO + OAB: **single terminal IDC** predicts storage-symptom resolution after TURP. NDO: higher Pdetmax, longer IDCs → upper-tract deterioration.
- **Cough-associated (stress-induced) DO**: mimics SUI but detrusor in origin.

Compliance

- P3 Urinary bladder compliance — *ΔV/ΔPdet; < 20 mL/cmH_2O poor; absolute pressure matters more; causes; pitfalls (fast fill, low-amplitude IDC, VUR/diverticulum/outlet pop-off)*
- Storage pressure **> 40 cmH_2O** → upper-tract damage. Outlet pop-off may appear only after sling/AUS → occlude outlet.

6 · Leak point pressures

- P1 Leak point pressures (VLPP, ALPP, DLPP) in SUI: clinical implications — *ALPP (low = weak sphincter) vs DLPP (high = upper-tract danger); technique from 150 mL, catheter out at capacity; VLPP < cough LPP; McGuire DLPP > 40 → 85% hydronephrosis/VUR*

ALPP (cmH_2O)	Inference (McGuire)
< 60	Intrinsic sphincter deficiency (ISD)
60–90	Equivocal
≥ 90	Little/no ISD (hypermobility)

- ISD and hypermobility (Q-tip > 30°) coexist; no hypermobility → SUI must be ISD; ALPP poorly predicts surgical success. Post-prostatectomy men often leak only with catheter out. ICS umbrella term: **urodynamic stress test**.
- **DLPP > 40 cmH_2O**: not a validated safe cut-off (20 more sensitive in children) — aim **as low as achievable**; lower bladder pressure (or outlet, e.g. sphincterotomy for DESD) and repeat UDS.
- **Occult SUI**: continent woman with POP leaks only on prolapse reduction.

7 · Urethral pressure profilometry (UPP)

- Brown–Wickham perfused catheter withdrawal. **MUCP** = max urethral − intravesical; women: functional length ~3 cm, MUCP 40–60 cmH_2O.
- **MUCP ≤ 20 cmH_2O** → ISD ("type III" SUI), predicted sling/suspension failure.
- ICS: unclear utility; EAU: do not use UPP or LPP to grade severity.

8 · Pressure–flow studies (PFS)

Pdet	Flow	Interpretation
Low/normal	High/normal	Unobstructed
High	Low	**BOO**
Low	Low	Detrusor underactivity (obstruction may be hidden)

- ICS: **detrusor underactivity** (reduced strength/duration → prolonged or incomplete emptying); **acontractile detrusor** ("neurogenic acontractile" replaces "areflexia").

Men — nomograms and indices

Abrams–Griffiths, Schäfer LinPURR (grades obstruction and contractility), ICS provisional nomogram — all from TURP patients.

Index	Formula	Cut-offs
BOOI	PdetQmax − 2Qmax	**> 40 obstructed · 20–40 equivocal · < 20 unobstructed**
BCI	PdetQmax + 5Qmax	**> 150 strong · 100–150 normal · < 100 weak**

- BOOI ≥ 40 needs Pdet ≥ 40 even at zero flow → decompensated detrusor masks obstruction.
- **Stop tests** (Piso): men Piso < 50 cmH_2O suggests underactivity.
- ~2/3 of men with BPO have DO; resolves in half to two-thirds after relief.
- **Micturitional UPP** (Yalla): drop > 5 cmH_2O within prostatic urethra localises obstruction.
- MCQ Bethanechol supersensitivity test (Lapides; Cannon's law): 2.5 mg SC; rise **≥ 15 cmH_2O** at 100 mL = positive (neurogenic); unreliable.

Women

- Women void at low pressure → **BOOI/ICS nomogram cannot be used**. Suspect with incomplete emptying, recurrent UTI, prior anti-incontinence surgery, prolapse.

Method	Criterion	Comment
Cut-points	e.g. Qmax ≤ 12 mL/s + PdetQmax ≥ 25 cmH_2O (Defreitas); Chassagne ≤ 15 / ≥ 20	Anatomic obstruction only; overlap
VUDS (Nitti)	Radiographic obstruction between bladder neck and meatus with sustained contraction of any magnitude	Localises; works despite weak detrusor
Blaivas–Groutz	Free Qmax vs Pdetmax; none/mild/moderate/severe	Two separate voids; **overdiagnoses**
Solomon–Greenwell	PdetQmax − 2.2Qmax: < 0 unlikely; > 18 almost certain	Includes functional obstruction

9 · EMG and sphincter coordination

- ICS standard: perineal surface electrodes; concentric needle better signal, uncomfortable.
- Normal: EMG quietens and urethral pressure falls **before** Pdet rises. **Guarding reflex**: EMG rises during an IDC — normal, not dyssynergia.

	DESD	Dysfunctional voiding
Cause	**Suprasacral cord lesion** (higher → likelier)	Learned; no neurological lesion
Behaviour	Involuntary sphincter contraction during DO/voiding	Intermittent sphincter contraction during voiding
Significance	High pressure → poor compliance → upper-tract risk	LUTS; fluoroscopy improves accuracy

- **DISD**: suprasacral lesion above T10–L1 sympathetic outflow; **VUDS only**. **PBNO**: bladder neck invisible to EMG → fluoroscopy.

10 · Videourodynamics

- P4 Video Urodynamics — *gold standard in NLUTD (EAU); only test for bladder neck/DISD/PBNO (types I–III); VUR pop-off; Nitti criteria; technique, screen only at key moments; indications*
- If VUDS unavailable in NLUTD: CMG + PFS (EAU).

11 · Ambulatory urodynamics

- **Natural filling**: some "poorly compliant" neuropathic bladders normal — rapid filling overstates pressure; DO detected more often.
- †Indication: symptoms unexplained by repeated conventional UDS. Penile cuff, wall thickness, NIRS — none replaces cystometry.

12 · Indications by condition (AUA/SUFU, EAU)

Female SUI

- Uncomplicated, demonstrated SUI ("index"): UDS may be omitted (AUA/SUFU 2023; EAU: not routinely).
- UDS in non-index: prior anti-incontinence/POP surgery, symptom–examination mismatch, voiding dysfunction, significant UUI, raised PVR, unconfirmed SUI, NLUTD. SUI not shown → repeat with catheter out; high-grade POP → reduce.

Male LUTS

- No UDS needed to start an α-blocker. AUA/SUFU: **PFS when obstruction matters, before invasive, morbid or irreversible treatment**; no obstruction → worse surgical results.
- EAU PFS triggers before invasive treatment: prior failed treatment; cannot void > 150 mL; **Qmax > 10 mL/s**; PVR > 300 mL; age > 80 or < 50.
- **Impaired compliance is the only absolute urodynamic indication to treat BOO.**
- OAB: no UDS initially; consider before invasive therapy.

Women with voiding symptoms; NLUTD

- PFS separates obstruction from impaired contractility; functional obstruction (PBNO, dysfunctional voiding) → VUDS.
- NLUTD goal: **low-pressure filling** even at cost of catheterisation; only UDS separates DO from poor compliance. Essential before treatment and in follow-up in SCI, myelomeningocele; helpful in BPO with Parkinson/MS; not usually first in post-stroke UUI.

13 · Landmark trials

Trial	Design	Result	Take-home
ValUE	RCT, 630 women, uncomplicated SUI: office evaluation ± UDS	Success ~77% both arms	Office evaluation alone non-inferior before SUI surgery
Dutch embedded RCT	Discordant UDS randomised: immediate mid-urethral sling vs UDS-tailored	Equal cure	UDS discordance need not change surgery
SISTEr	Burch vs pubovaginal sling	ALPP/DO did not predict outcome	UDS adds little in stress-predominant SUI

109 Neuromuscular Dysfunction of the Lower Urinary Tract

1 · Principles and classification

- Neuropathic lower urinary tract (LUT) dysfunction is largely **predictable from lesion site**; acute may differ from chronic.
- **Five ICS parameters**: sensation; detrusor (normal, overactive, areflexic, impaired); compliance; smooth sphincter (synergic/dyssynergic); striated sphincter (synergic, dyssynergic, bradykinetic, impaired voluntary control, fixed tone).
- **Neuroplasticity**: reversible, then fixed. In bladder outlet obstruction (BOO): ↑ type III:I collagen (↓ compliance), denervation, nerve growth factor (NGF) afferent sensitisation → detrusor overactivity (DO) persisting after de-obstruction.

Level	Typical disorders	Detrusor	Smooth sphincter	Striated sphincter
Suprapontine	Stroke, tumour, Parkinson disease (PD), dementia, cerebral palsy	Overactive	Synergic	Synergic (± impaired voluntary control; PD bradykinesia)
Multiple system atrophy (MSA)	—	Overactive → impaired	**Open**	Denervated
Suprasacral cord	Spinal cord injury (SCI), multiple sclerosis (MS), transverse myelitis	Overactive	Synergic (dyssynergic if above T7)	**Dyssynergic**
Sacral cord / cauda equina	Sacral SCI, disc, myelodysplasia	Areflexic	Competent non-relaxing; may open	Fixed tone
Peripheral	Radical pelvic surgery, diabetes, tabes	Impaired / areflexic	Open (pelvic surgery)	Fixed tone / synergic

KEY POINT

Dyssynergia appears only when the lesion lies **between the pons and the sacral cord**; an open bladder neck marks autonomic (MSA, radical pelvic surgery, myelodysplasia) or sacral damage. PD and MSA share the detrusor pattern but **not the outlet**.

2 · Lesions at or above the brainstem

Stroke

- Urinary incontinence (UI) persisting 7 days predicts poor survival; commonest = impaired-awareness UI > urgency UI (UUI).
- **Acute**: retention ("cerebral shock") → **chronic**: DO, intact sensation, **synergic sphincters**; UUI if voluntary sphincter control lost (cortex/internal capsule).
- **True detrusor–sphincter dyssynergia (DSD) does not occur**: **pseudodyssynergia** = electromyographic (EMG) flare from voluntary attempts to inhibit a contraction.
- Men with LUT symptoms (LUTS): full urodynamic study (UDS) before outlet surgery to exclude DO with impaired contractility.

Dementia, brain injury, tumour, others

- **Dementia**: anticholinergics worsen cognition, falls, dementia risk → **β3 agonist strongly preferred** (SUFU).
- **Brain injury**: above pontine micturition centre (PMC) → DO, synergic; brainstem below PMC → DSD possible.
- **Brain tumour**: site (superior frontal lobe), not histology; frontal lesions can cause retention.
- **Normal-pressure hydrocephalus**: dementia + gait ataxia + UI; shunt improves UUI.
- **Cerebral palsy**: DO, synergic; adults with LUTS — poor compliance/detrusor leak point pressure (DLPP) ≥40 cm H_2O in ~half → full evaluation; conservative care succeeds in ~¾.

Parkinson disease vs multiple system atrophy

- **PD**: nigral dopamine loss (loss of inhibitory D1, unopposed D2). LUTS 35–70%, storage dominant; sphincter **bradykinesia** mimics DSD.
- Treatment: low-CNS overactive bladder (OAB) drugs (mirabegron, trospium, darifenacin, ER fesoterodine); onabotulinumtoxinA; PTNS and SNM effective; subthalamic deep brain stimulation improves capacity.
- **TURP in PD not contraindicated** — old "post-TURP incontinence in PD" cases were MSA. Confirm PD, exclude MSA, videourodynamics first.
- **MSA**: glial α-synucleinopathy; loss in intermediolateral columns and **Onuf nucleus** → early severe LUTS + erectile dysfunction (ED); DO early, poor emptying late. Treating DO worsens emptying → add clean intermittent catheterisation (CIC); desmopressin for nocturia. **Prostatectomy hazardous.**

Feature	MSA	PD
Urinary symptoms / ED	Precede or accompany parkinsonism; ED near universal	Years after diagnosis; ED ~⅓
Dominant problem; PVR	UI; high PVR	Frequency/urgency; PVR normal
Bladder neck	**Open**	Synergic
Striated sphincter EMG	**Denervation**	Normal (bradykinesia)
Detrusor	DO → areflexia	DO; areflexia uncommon
Outlet surgery	Worsens	Acceptable after work-up

3 · Spinal cord disease

Multiple sclerosis

- LUTS 50–90%; presenting complaint in up to 15%. UDS: DO ~⅔; DSD in 30–65% of those with DO; areflexia ~¼.
- Symptoms do not track disease → **baseline UDS essential**.
- Management **reversible** (pattern changes): antimuscarinic + CIC; onabotulinumtoxinA; SNM in selected (MRI-compatible). Credé/Valsalva contraindicated; indwelling catheter last resort, suprapubic if used.

Spinal cord injury

- P2 Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male — *spinal shock, suprasacral NDO + DSD, autonomic dysreflexia,*

CIC/toxin/sphincterotomy/Brindley ladder

- Sacral cord begins at column T12–L1; conus → cauda equina ~L2. Stone risk highest first 3 months.
- Spinal shock usually **6–12 weeks**; reflexes return with lower-limb deep tendon reflexes. Smooth sphincter dyssynergia with cord **T7–T8** (column T6) or above.
- **Sacral**: areflexia, compliance normal then may fall (decentralisation); non-relaxing bladder neck (may later open); fixed striated tone. Silent upper-tract damage possible **without VUR**.

	Suprasacral (complete)	Sacral (complete)
Detrusor	Overactive	Areflexic
Compliance	Normal initially	Normal/high → may fall
Smooth sphincter	Synergic (dyssynergic above T7–T8)	Competent non-relaxing → may open
Striated sphincter	Dyssynergic	Fixed tone, no voluntary control
Somatic signs	Spasticity, hyperreflexia	Flaccidity, ↓ reflexes
Management	Lower pressure (drugs/toxin) + CIC	CIC + keep compliance safe
Management	Lower pressure (drugs/toxin) + CIC	CIC + keep compliance safe

- **Neurology ≠ urodynamics**: occult levels; thoracolumbar (T10–L2 column) lesions least predictable. **Manage by UDS, not level**; one study may not suffice.
- Upper-tract risk: detrusor fibrosis, leak point pressure ≥75 cm H_2O, DO for >⅓ of filling. †DLPP >40 cm H_2O = classic threshold.

Autonomic dysreflexia (AD)

- SCI **above T6**; may appear years later. Triggers: bladder/rectal distension, blocked catheter, instrumentation, stones, sores, fracture, sex.
- UDS risk: complete, cervical, DSD, poor compliance.

> **ALGORITHM**
> †Sit upright, loosen clothing → **remove stimulus** (drain bladder, unblock catheter, rectum) → persistent: nitroglycerin 2% paste above lesion (first line) → labetalol / IV nitroprusside → prophylaxis terazosin or prazosin (α1) → intractable: sacral rhizotomy, sympathectomy, cordectomy

- Nifedipine (10–20 mg sublingual) now banned in many centres — uncontrolled hypotension.

VUR and women with SCI

- **Chronic urethral catheter worst** for upper tracts. VUR artefactually enlarges capacity and lowers pressures on UDS.
- Destroyed urethra: bladder neck closure + suprapubic catheter, or diversion.
- Women: no collecting device — paraplegics CIC, tetraplegics often indwelling.

Other cord disorders

- **Transverse myelitis**: retention → DO + DSD; manage as SCI.
- **Cervical myelopathy**: DO ± DSD, or underactivity (posterior funiculus) → UDS guides.
- **Neurospinal dysraphism**: myelomeningocele ~90% of spina bifida cystica; level correlates poorly with UDS. Typical adult: **areflexic bladder, open bladder neck**, fixed sphincter → leaks at fixed pressure; DSD 10–15%. Poor compliance, high pressure, DO predict renal

decline. Stress UI: bulking safer than sling/AUS (does not reduce compliance). Medical + toxin → augmentation/diversion.

- **Tethered cord**: after growth spurts; DO commonest; urinary recovery lags pain/motor.
- **Sensory neurogenic bladder** (tabes, B12): large capacity. **Motor** (polio): retention, intact sensation.

4 · Lesions distal to the cord

- **Disc disease** (L4–L5, L5–S1): commonest **normally compliant areflexic bladder** (preganglionic injury incomplete). **Pre-laminectomy UDS** (clinical and medicolegal).
- **Cauda equina syndrome**: saddle anaesthesia, loss of voluntary sphincter control and sexual response; acontractile detrusor. **Decompression within 48 h** → better urinary/rectal recovery. Sphincter recovers before detrusor.
- **Radical pelvic surgery** (APR, radical hysterectomy): impaired contractility + fixed striated tone + **open non-functional bladder neck** + ↓ compliance → leak or retention; positive bethanechol supersensitivity. **CIC and wait 6–12 months**; **no prostatectomy unless BOO proven**.
- **Herpes zoster**: retention after rash; resolves 1–2 months. **Herpes simplex** sacral radiculitis + retention + CSF pleocytosis = **Elsberg syndrome**, transient.
- P2 Bladder dysfunction non genitourinary systemic disease — *diabetic cystopathy (↓ sensation → ↑ capacity → ↓ contractility → ↑ PVR; DO common; pressure–flow vs BOO; timed voiding), Guillain–Barré, lupus, B12*
- Diabetic "dyssynergia" is straining artefact. **Guillain–Barré**: retention → CIC pending recovery.

> **KEY POINT**
> Distal lesions hit the peripheral reflex arc → areflexia with a non-relaxing/fixed outlet; compliance falls only when preganglionic parasympathetic motor fibres are damaged. Hence urgent decompression, waiting and CIC outrank outlet surgery.

5 · Specific functional and miscellaneous conditions

Detrusor–sphincter dyssynergia

- P1 Detrusor–external sphincter dyssynergia (DESD / DSD) — *lesion pons–sacral cord, Blaivas types 1–3, pseudodyssynergia, half of untreated men deteriorate, CIC first, sphincterotomy if CIC impossible*
- No lesion → doubt DSD (exception: Hinman syndrome in children). Misdiagnosed pseudodyssynergia → harmful treatment.
- Blaivas type 1 (incomplete): observe unless VUR, hydronephrosis or AD; types 2–3: treat.

Functional outlet disorders

- **Dysfunctional voiding** (Hinman): striated contraction during voiding without neurological disease, proven on pressure–flow–EMG.
- **Non-neurogenic chronic retention**: PVR ≥ **300 mL** persisting ≥6 months on ≥2 occasions. High risk (hydronephrosis, CKD, UTI) → drainage; suprapubic > urethral long term.
- **Primary bladder neck obstruction**: young/middle-aged men, normal prostate; videourodynamics localises BOO to bladder neck. α-blockers help some; **bladder neck incision definitive**. "Trapped prostate" (Turner-Warwick) → small TURP + bladder neck.

- **Female BOO**: rare; Groutz criteria — free Qmax ≤12 mL/s + Pdet at Qmax ≥20 cm H_2O. Bladder neck surgery risks sphincteric UI.
- **Fowler syndrome**: retention in young women (capacity >1 L), polycystic ovaries; needle EMG **complex repetitive discharges and decelerating bursts**; urethral closure pressure >100 cm H_2O. **SNM is the treatment** (~70%); Fowler predicts SNM response.

Other conditions

- **Myasthenia**: incontinence after prostatectomy. **Isaacs**: sphincter spasm → suprapubic drainage, plasmapheresis. **Wernicke**: thiamine.
- Myelopathies (tropical spastic, schistosomal, syringomyelia) → DO + DSD; CMV, Lyme → areflexia. **Lupus cystitis** → steroids.
- **Radiation**: early storage symptoms recover by 6 months. **Defunctionalised bladder**: cycling; transplant without augmentation.

6 · Evaluation and risk stratification

- Guidelines: AUA/SUFU neurogenic LUT dysfunction (NLUTD) 2021; EAU Neuro-urology 2022. **Primary goal: prevent renal damage.**
- All: history, examination (cognition, **hand dexterity**, spasticity), urinalysis, PVR, diaries.
- **Defer full assessment during spinal shock or acute brain injury**; no irreversible therapy then.

Risk (AUA/SUFU)	Typical	Defining findings
Low (all)	Suprapontine or uncomplicated peripheral	Spontaneous voiding, low PVR, stable; normal imaging/renal function
Unknown	Infrapontine–suprasacral; sacral/infrasacral	Storage + voiding symptoms, ↑ PVR, recurrent UTI; not yet assessed → UDS, upper-tract imaging, renal function
Moderate	—	Neurogenic retention; DO with incomplete emptying; normal tracts
High (any one)	—	Poor compliance, ↑ storage pressure, DO with DSD, VUR; hydroureteronephrosis, scarring, large stones; abnormal renal function

KEY POINT

Low risk needs no routine imaging, renal function or UDS. DO alone fits low risk (suprapontine, synergic); high risk means DO with infrapontine pressure consequences. Risk is judged by **pressure, compliance and upper tracts**, not diagnosis.

7 · Management

- **Goals**: protect upper tracts; control infection; low-pressure storage and emptying; continence; avoid catheter/stoma if possible; social acceptability. Start simplest, most reversible.

ALGORITHM

Low-pressure storage + regular emptying: CIC + antimuscarinic / β3 agonist → intradetrusor onabotulinumtoxinA → neuromodulation (selected) → augmentation / continent diversion → bladder neck closure + suprapubic catheter / ileovesicostomy / incontinent diversion (last resort)

- **CIC gold standard**; volumes < **500 mL**; Credé/Valsalva not recommended. CIC beats indwelling urethral catheter (less pyelonephritis, epididymo-orchitis, urosepsis); chronic catheter → **suprapubic > urethral**.
- Refractory NDO → **intradetrusor onabotulinumtoxinA** (~60% continue long term); warn about CIC. Intrasphincteric toxin not routine.
- Brindley (deafferentation + anterior root stimulator) in complete SCI: rhizotomy benefits even if stimulator off *→ Ch 115*. PTNS for voiding patients with OAB (MS, PD, stroke); SNM for NLUTD without SCI or spina bifida.
- Bladder neck closure: retropubic better than perineal.
- **UTI**: treat bacteriuria only if symptomatic; correct stones/residuals; **no prophylaxis** for CIC or indwelling catheters. Refractory → upper-tract imaging, cystoscopy, UDS.
- **Bladder cancer**: incidence comparable to general population but often muscle-invasive at diagnosis; catheters and UTI are the risk. **AUA/SUFU 2021: no routine surveillance cystoscopy**; scope for haematuria.

8 · Follow-up

Body	Schedule
AUA/SUFU 2021 (risk-based)	Low risk: none routine. Moderate/high: yearly history, examination, renal function; upper-tract imaging every 1–2 years; repeat UDS for change; bowel reconstruction: yearly electrolytes/acid–base, imaging
EAU	Specialist review every 1–2 years; ultrasound + residual 6-monthly; dipstick 2-monthly; yearly examination and bloods
American Paraplegia Society (SCI)	Annual for 5–10 years then alternate years; renogram — >20% fall in renal plasma flow → investigate; annual cystoscopy with indwelling catheter (**not endorsed** by AUA/SUFU)

110 Overactive Bladder

1 · Definitions and terminology

- P1 P2 Describe pathophysiology of OAB. Briefly outline management &describe nonmedical non-pharmacological management; Overactive bladder and its drug therapy — *ICS definition, OAB vs DO, afferent/urothelial/myogenic/central mechanisms, behavioural therapy, drugs, PTNS/toxin/SNM*
- Overactive bladder (OAB): **urgency is the defining symptom**; "urge" became "urgency" incontinence to tie it to that sensation. OAB = symptom syndrome; detrusor overactivity (DO) = urodynamic observation, may be asymptomatic or artefact.

Term	ICS definition (all patient-reported)
Increased daytime frequency	Patient considers they void too often by day; upper normal **7** voids
Nocturia	Waking ≥1 time to void; each void preceded and followed by sleep
Urgency	Sudden, compelling desire to void, difficult to defer
UUI	Involuntary leakage associated with urgency

- **Bother**: **urgency urinary incontinence (UUI) is the most bothersome LUTS in both sexes**. Frequency bothers at ≥**8** daytime voids; nocturia at ≥**2**.
- **Isolated nocturia** usually = nocturnal polyuria or poor sleep. †Nocturnal polyuria (ICS): night urine >33% of 24-h output in older adults.
- **Mixed urinary incontinence (MUI)** = stress UI (SUI) + UUI. SUI + "OAB dry" is **not** MUI. Urinary + faecal = **double incontinence**. OAB wet: more psychological burden, more DO (almost always in men).

OAB versus bladder pain syndrome (BPS)

BPS requires bladder pain/pressure/discomfort + ≥1 urinary symptom; pain excluded from OAB. Overlap large (shared afferent hypersensitivity).

Feature	OAB	BPS
Primary fear	**Leakage**	**Pain**
Filling sensation	Phasic urgency waves	Steady, accelerating pain
Voided volumes	Variable	More consistent
Deferral	Limited — leak risk	Possible, at cost of pain
Site	Perineum/penile base, urethra/vagina	Suprapubic ± perineal
UUI	Yes	No

2 · Epidemiology and burden

- Adult OAB ≈ **12–29%**: loosest definition (EpiLUTS) highest, strict ICS (EPIC) lowest.
- Women: OAB 1.5–2×, UUI 2–3× men (NOBLE UUI 7.6% vs 2.6%) — sex gap is in **wet** OAB. Beyond 75: ~40% of men, ~⅓ of women. Highest in Black people.
- OAB dry → UUI ~1%/year; remission ~6%/year.
- <Half tell a clinician. ~**92%** stop the first antimuscarinic within 2 years; reasons: **unmet expectations** > side effects > cost.
- Third-line: onabotulinumtoxinA 100 U most cost-effective; sacral nerve stimulation (SNS) costliest.

3 · Pathophysiology

- **DO no longer the sine qua non**: only ~half with DO have urgency; **44–69%** with urgency show DO.
- **Periaqueductal grey (PAG) + pontine micturition centre (PMC) = main regulatory level**; forebrain (insula, prefrontal) — sensation, volition.
- Silent C fibres awaken; urothelium-based vs myogenic (exaggerated **micromotions**) hypotheses; bowel cross-talk; failed CNS filtering of "afferent noise" — see pointer.

> **KEY POINT**
> Final common pathway: more afferent signal reaches awareness — the bladder or a neighbour turns it up, or the CNS fails to turn it down. Hence DO and obstruction correlate only loosely with OAB.

4 · Aetiology and risk factors

- **Ageing** (clearest), female sex, metabolic syndrome, bowel, hormones; depression/anxiety.
- **BOO**: standard animal model, but **indirect** in humans — OAB similar in women (BOO rare); probably parallel consequences of ageing.
- **Behaviour**: "just in case" voiding; coping (toilet mapping, fluid restriction) sustains negative bladder perception.

5 · Clinical assessment

- **Minimum** (AUA/SUFU; EAU concurs): **history, examination, urinalysis** → empirical treatment. EAU adds questionnaires and diary.
- **History**: bother; SUI; voiding symptoms; haematuria; fluids/caffeine; occult neurology (new OAB with ED or tremor); pelvic surgery/radiotherapy; bowel (faecal incontinence ~**50%** of women with OAB); drugs; polyuria (diabetes, sleep apnoea, lithium).

Drug class	Relative contraindications
Antimuscarinics	Uncontrolled closed-angle glaucoma; cognitive impairment; retention; impaired gastric emptying
β3-adrenergic agonists	Uncontrolled hypertension; arrhythmia

- **Examination**: frailty (Timed Up and Go), cognition, leg oedema. **Pelvic/genital examination most important** — prolapse, atrophy, stress test; DRE in men; perineal sensation, anal tone.
- **Urinalysis mandatory**; culture if infection suspected; microhaematuria → full evaluation.
- **PVR**: not needed before conservative/oral therapy if uncomplicated; measure if retention risk, voiding symptoms (men), prior outlet surgery, neurology.
- **Diary** 3–7 days (minimum 24–48 h). **Maximum voided volume = functional capacity**. Sensation grade 0 = "just in case" void; 2–3 urgency; 4 urgency with leak.
- PROMs: ICIQ-OAB + ICIQ-OABqol; AUA/SUFU: UDI-6, IIQ or OAB-q. MCQ OAB-V8 = screening tool, not severity score.

6 · Urodynamics (UDS)

- **Not** in initial work-up (nor cystoscopy, ultrasound). **Indicated** when result decides invasive/irreversible treatment, after multiple failures, or suspected BOO (OAB after outlet procedure). Index woman: improves satisfaction, not outcome.
- Findings: **DO** or **increased sensation**. Stop antimuscarinics to show DO; continue to explain refractory symptoms.
- DO in OAB dry: 60–70% men, 40–50% women; OAB wet: **>90% men**, absent in 30–40% of women (SUI misread as urgency). DO correlates **best with UUI, worst with nocturia**.
- Provoke (running water, standing); low-volume DO masking SUI → refill supine.
- **Assess emptying**: each **10 mL** PVR raises catheterisation risk after onabotulinumtoxinA by **27%**; DO persists in up to **60%** after TURP.

7 · Management

- Benign → **no treatment is acceptable**; incontinence-associated dermatitis justifies more aggressive care. Immobility-driven leakage needs mobility treatment.
- **Realistic expectations** — improvement, not cure. AUA/SUFU lines **not compulsory sequence**; allow **4–12 weeks** per therapy.

> **ALGORITHM**
> Confirm OAB, exclude mimics → education → conservative → drugs → tibial nerve stimulation / sacral neuromodulation / onabotulinumtoxinA → augmentation or diversion; reassess at every step

- Conservative: fluid ↓ ~**25%**; caffeine ↓ improves urgency/frequency, not consistently UUI; PFMT + bladder training: leaks ↓ **50–80%**.
- **Weight loss** reduces SUI and UUI (†PRIDE randomised trial, obese women); bariatric surgery: each 5% lost → ~7–8% lower incontinence risk.
- Drugs *→ Ch 113*: response idiosyncratic — switch or combine before invasive work-up. **Dementia**: antimuscarinic >**3 months** → 10–65% higher risk → β3 agonist. BPH + OAB: α1-blocker + antimuscarinic; retention ~3% → caution if PVR raised.

Third-line *→ Ch 115*

Frailty predicts functional decline and 1-year morbidity up to 50% even after minor procedures.

	Tibial nerve stimulation	OnabotulinumtoxinA	SNS
Efficacy	60–80% respond	UUI ↓59%; ~¼ dry	5-year success 70–80%
Drawback	12 weekly visits; response by weeks 8–10	UTI ~20%; catheterisation 0–12% in trials, real-world ~20% women, ≤35% men	Revision >30% at 5 years; battery
Risk	Lowest	Moderate	Highest

- **ICI pathways**: men — treat outlet alongside bladder (α-blocker if BOO); women — most bothersome MUI component first; frail elderly — comorbidity and carer goals first, cautious antimuscarinic trial.

KEY POINT

Escalate by invasiveness and reversibility, not severity alone: OAB is benign, so patient goals, frailty and PVR decide how far to climb.

111 Underactive Detrusor

- P2 P4 Management of detrusor underactivity; Detrussor underactivity. — *ICS definition, causes (neurogenic/myogenic/drugs/functional), BCI, management table: CISC mainstay, α-blockers, bethanechol not routine, SNM, Brindley, outlet surgery, diversion, investigational options*

1 · Definitions and symptoms

- Drugs and surgery lower outlet resistance but cannot strengthen the detrusor → **detrusor underactivity (DUA)** management is largely drainage (intermittent self-catheterisation, ISC).
- **DUA** (ICS 2002, urodynamic): contraction of **reduced strength and/or duration** → prolonged/incomplete emptying. **No thresholds** specified.
- **Acontractile detrusor**: no contraction. **"Bashful bladder"**: fails to void during UDS only, often cannot void in public toilets.
- **Underactive bladder (UAB)** (ICS 2018, symptomatic): slow stream, hesitancy, straining ± incomplete emptying, sometimes storage symptoms. **No pathognomonic symptom**; overlaps OAB and BOO.
- Significant postvoid residual (PVR) not agreed; proposal >**40%** of functional capacity (voided volume + PVR).

Phase	Symptoms
Voiding	Weak stream, intermittency, hesitancy, straining
Post-micturition	Incomplete emptying
Storage	Frequency and nocturia, **or** reduced desire with infrequent voiding
Very large PVR	Incontinence, especially in sleep

KEY POINT

Sensation decides the picture. Absent sensation → infrequent voider; poor emptying → overflow worse at night (urethra relaxes in sleep). Preserved sensation → frequent voider with incomplete emptying and nocturia. DUA may be asymptomatic.

2 · Epidemiology

- Population prevalence **unknown** (no non-invasive marker). In women BOO rare (2.7–8% of referrals), so low flow + high PVR suggests DUA.
- UDS series: men <50 **9–28%**; men >70 ~**48%**; women 12–45%; highest in institutionalised elderly, often **detrusor hyperactivity with impaired contractility (DHIC)** — involuntary contraction emptying <half the bladder.
- Coexisting dysfunction in elderly: ~half of men DO or BOO; ~¾ of women DO or SUI → **UDS before LUTS surgery in the elderly**.
- Non-neurogenic male DUA over 10 years: **minimal progression**.

3 · Aetiopathogenesis

Confirmed: neurological disease and diabetes. Probable: BOO, ageing, atherosclerosis.

Mechanism	Site	Result
Myogenic	Myocyte, extracellular matrix (ECM)	Weak contraction despite normal drive

Mechanism	Site	Result
Efferent	Sacral cord, roots, pelvic nerves	Impaired activation
Afferent	Peripheral afferents, spinal columns	**Early termination** of the voiding reflex
Brain	PMC, PAG, limbic, prefrontal	Failed integration

- **Myogenic**: ECM fibrosis; Elbadawi **degeneration pattern** on electron microscopy. Same fibrosis in asymptomatic elderly women → ageing contributes.
- **Afferent**: bladder (volume, contraction) and urethral (flow) afferents sustain the reflex; sensation falls with age.
- **"Stiff balloon" hypothesis (Smith)**: fibrosis → abrupt wall-stress rise → late sensation then urgency → small void → stress collapses → afferent drive fades → contraction stops early → PVR.
- Causes also include tabes, spinal stenosis, pelvic fracture, radiotherapy; **drugs reversible** (anticholinergics incl. antihistamines, antipsychotics, tricyclics, antiparkinsonian; **opioids**).
- †**Fowler syndrome**: young women in retention, non-relaxing striated sphincter (abnormal EMG); responds to sacral neuromodulation.
- **BOO**: animal model hypertrophy → **decompensation**; proposed Laplace wall tension → ischaemia–reperfusion → denervation. But in men followed >10 years Pdet@Qmax did not fall; most men in acute retention have preserved detrusor function.
- **Diabetes — temporal theory**: early osmotic diuresis → hypertrophy, storage symptoms; late oxidative damage → poor sensation and emptying.

Condition	DUA	Key point
Stroke	~Half retain acutely	"Cerebral shock"; chronic sequel is DO
Parkinson disease	<20%	DO commoner; DUA drug-related
Multisystem atrophy	**52–95%**	Efferent parasympathetic atrophy; mistaken for Parkinson
Multiple sclerosis	~20%	Lumbosacral plaques
Radical pelvic surgery	Hysterectomy: ~72% LUT dysfunction	Nerve-sparing reduces; many recover by 1 year
Guillain–Barré, zoster	—	Reversible

4 · Diagnosis

- **Invasive UDS only accepted method**; no consensus criteria; nearly all measure strength, not speed/duration.
- Low Qmax + low Pdet@Qmax. Pitfalls: **bladder outlet relation** (Hill equation — pressure and flow inverse) → Pdet@Qmax underestimates strength; outlet resistance ignored. Hence isovolumetric estimates.

Measure	Formula	Thresholds / note
Bladder contractility index (BCI) (Abrams)	Pdet@Qmax + 5Qmax	>150 strong; 100–150 normal; **<100 weak**; male-derived
Projected isovolumetric pressure (PIP) (Schäfer)	Pdet@Qmax + K·Qmax (K 5 men, 1 older women)	<50 very weak; 50–100 weak

Measure	Formula	Thresholds / note
Detrusor contraction coefficient (DECO)	BCI/100	<1 weak

- †**BOO index** = Pdet@Qmax − 2Qmax; >40 obstructed, <20 unobstructed.
- **Stop tests**: voluntary reads ~20 cm H_2O lower (reflex inhibition); continuous occlusion most reliable but painful — research only.
- Speed: fall in shortening velocity precedes fall in Watts factor (early marker).
- Failed void on UDS → **ambulatory UDS**: >⅘ of apparently acontractile patients contract.

5 · Management

ALGORITHM

Evaluate (diary, DRE, urinalysis, flow, PVR; neurological deficit → spinal MRI; stop offending drugs; treat constipation) → timed/double voiding → lower outlet resistance where relevant → ISC for high PVR (suprapubic if ISC impossible) → selected: sacral neuromodulation, outlet surgery, diversion

Aim	Routine	Experimental	Not recommended
Drainage	Timed/double voiding; catheterisation; diversion	—	Credé, Valsalva, reflex voiding
↓ Outlet resistance	Pelvic floor biofeedback; α-blockers; outlet surgery	Sphincter botulinum toxin	—
↑ Detrusor function	Sacral neuromodulation; anterior root stimulator	Intravesical prostanoids; intravesical electrotherapy; myoplasty	**Muscarinic agonists; anticholinesterases**

- **Credé/Valsalva** only with incompetent sphincter (neurogenic); otherwise reflux into ureters and prostate.
- **ISC**: urethral bleeding ~⅓; †keep volumes below ~500 mL. Suprapubic best long-term indwelling.
- **Bethanechol, distigmine**: bronchospasm, diarrhoea, salivation, **rare fatal cardiac arrest**; useless with fibrosis. α-blocker + parasympathomimetic improved flow in one single-blind RCT. PGE_2 (contracts detrusor, relaxes urethra), ASP8302, TAC-302 investigational → *Ch 113*.
- **SNM**: inhibits urethral afferents that block bladder afferents; best with sphincter/pelvic floor disorder; potential in DHIC (treating DO worsens DUA and vice versa) → *Ch 115*.
- **Brindley**: complete SCI, needs efferents and muscle, with rhizotomy. **Intravesical electrotherapy (Katona)**: Aδ afferents; mainly children; only sham-controlled trial negative.

KEY POINT

Every therapy needs a surviving link: Brindley needs efferents and muscle; neuromodulation and electrotherapy need afferents and muscle. **Preserved sensation predicts response** (bethanechol, toxin, electrotherapy).

- **Sphincter toxin**: 50–100 U cystoscopic (men) or periurethral (women); ~half of idiopathic improve; best with poor relaxation, good sensation; short-lived, unlicensed.

	DUA, low PVR, symptomatic	DUA, catheter-dependent
Reasoning	Emptying at low Pdet → BOO unlikely significant	Lower resistance enough to void

	DUA, low PVR, symptomatic	DUA, catheter-dependent
Evidence	TURP at ~11 years: no benefit over conservative care	Flow rarely normalises, but many void
Verdict	**Surgery unlikely to help**	Consider in young, fit patients

- Poor outcome in retention: voiding pressure < **45** cm H_2O, age > **80**, PVR > **1500 mL**.
- Women: no bladder neck incision (incontinence, stenosis).
- **Diversion**: continent catheterisable stoma (young women, idiopathic retention refractory to SNM); ileal conduit for renal deterioration from intramural ureteric obstruction.
- Latissimus dorsi myoplasty: most acontractile patients voided; complications ⅓. Reduction cystoplasty abandoned.

113 Pharmacologic Management of Lower Urinary Tract Storage and Emptying Failure

1 · Framework

Wein functional scheme *→ Ch 104*: **storage failure** — relax bladder or ↑ outlet resistance; **emptying failure** — ↑ detrusor contraction (weak tools) or ↓ outlet resistance. Most LUT drugs fail on **selectivity/therapeutic window** — the same receptor in heart and vessels caps the dose.

2 · Antimuscarinics and β3-agonists

- P1 P3 Compare and contrast various detrusor relaxants in management of OAB; OAB Pharmacotherapy: Darifenacin vs Solifenacin vs Tolterodine, and Mirabegron — *M3 vs M2, agent table with doses, β3/cAMP, mirabegron ≥180/110 and CYP2D6 (metoprolol, desipramine, digoxin), vibegron 75 mg, dementia risk, SYNERGY/BESIDE*
- † Act mainly in **storage** (↓ urgency, ↑ capacity, damp afferents), not on voiding contraction → retention uncommon.
- Extra agent points: **oxybutynin** — highest dry mouth (N-desethyloxybutynin); transdermal bypasses first-pass → less dry mouth; most CNS penetration — avoid in elderly (Beers). Tolterodine 2 mg bd or ER 4 mg od. **Darifenacin** most M3-selective; constipation. **Trospium** quaternary amine, renally excreted. Propiverine 20 mg (mixed action); imidafenacin 0.1–0.2 mg bd (Japan).
- † **Side effects by receptor**: dry mouth, constipation, blurred vision (M3); tachycardia (M2); cognition (**M1**).
- † **Contraindications**: **uncontrolled narrow-angle glaucoma**, urinary/gastric retention, myasthenia gravis, severe ulcerative colitis/toxic megacolon.
- † Elderly: cumulative **anticholinergic burden** → β3-agonist or non-CNS-penetrant agent. Men with BOO: safe if PVR low; check PVR before and during.
- Mirabegron 25 → 50 mg od; avoid in severe uncontrolled hypertension (≥**180/110** mm Hg). Vibegron: no titration, no BP effect in trials.

ALGORITHM

† OAB: behavioural therapy → antimuscarinic *or* β3-agonist (β3 first in elderly/cognitive risk) → switch or combine → onabotA 100 U / SNM / PTNS → augmentation cystoplasty

3 · Combination therapy

Direction	Evidence	Agents
Mirabegron added to failing antimuscarinic	SYNERGY, SYNERGY II, BESIDE, MILAI	**Solifenacin only**
Antimuscarinic added to failing mirabegron	MILAI II (52 weeks)	Solifenacin 5, propiverine 20, imidafenacin 0.2, tolterodine ER 4 mg
Vibegron + antimuscarinic	Phase IIb (vibegron 100 + tolterodine 4 mg)	Extrapolated; no trial after monotherapy failure

KEY POINT

Combination beats solifenacin 5 mg and either monotherapy; in non-responders it **equals** solifenacin 10 mg with less dry mouth — **tolerability, not efficacy**, over dose escalation. No change in heart rate, BP, QT or PVR; constipation roughly doubles. Benefit irrespective of age ≥65 or ≥75.

- **α-blocker + desmopressin**: refractory nocturia — ≈30% of men ≥60 with BPH have nocturnal polyuria; evidence uncontrolled.
- **Vaginal oestrogen + antimuscarinic**: additive benefit **not established** (simultaneous RCTs null; vaginal estradiol ≈ oxybutynin). † Systemic oestrogen worsens UI — do not use.

4 · Intradetrusor onabotulinumtoxinA (onabotA)

- P1 Botulinum toxin in urology: mechanism, indications and doses — *SNAP-25 cleavage, 100 U/20 sites OAB (EMBARK), 200 U/30 sites NDO (DIGNITY), PVR at 2 weeks, UTI, myasthenia*
- Doses: **100 U** idiopathic OAB; **200 U** NDO. DIGNITY: equal efficacy in anticholinergic users and non-users; EMBARK: refractory only.
- † Retention/CIC higher in NDO non-catheterisers (MS) — must accept CIC first. Commonest AE UTI. Lasts ≈6–9 months; repeat ≥12 weeks apart. Caution aminoglycosides.

5 · Nocturia — desmopressin †

- P3 Desmopressin (DDAVP) — *V2 agonist for nocturnal polyuria; hyponatraemia risk age ≥65; sodium at baseline, 1 week, 1 month; contraindications incl. loop diuretics*
- Low-dose sublingual: women 25 µg, men 50 µg; key risk rises at age ≥**65**.

6 · Increasing outlet resistance (SUI)

† PFMT and surgery first-line; no drug standard. Systemic α-agonists lack urethral selectivity — BP rises before closure improves.

Drug	Mechanism	Status
Duloxetine †	SNRI at **Onuf's nucleus** → ↑ pudendal tone	40 mg bd; nausea → discontinuation; Europe, not USA
Phenylpropanolamine, midodrine, norepinephrine	Systemic α-agonists	Weak evidence; haemorrhagic stroke → phenylpropanolamine banned (FDA)
Imipramine †	Anticholinergic + α-like outlet effect	Mixed UI, enuresis; orthostatic hypotension, arrhythmia — caution elderly
Clenbuterol	β-agonist	Small flawed RCT; tremor; abandoned
PSD503 vaginal phenylephrine gel	Local α-agonist	Poor acceptability; abandoned

7 · Facilitating emptying

- † **Bethanechol**: **no convincing controlled evidence** in detrusor underactivity; cholinergic AEs. Contraindicated: asthma, peptic ulcer, bradycardia, mechanical obstruction.
- † **α-blockers**: BPH, primary bladder neck obstruction, neurogenic LUT dysfunction; aid trial without catheter.
- † Striated sphincter: baclofen, dantrolene — limited; sphincteric onabotA.

- † **Retention-causing drugs**: anticholinergics (antihistamines, tricyclics, antipsychotics, antiparkinsonian), pseudoephedrine, opioids, benzodiazepines, anaesthesia/epidurals — ask in every retention.

8 · Failed drugs

Agent	Indication	Reason for failure
Vitamin D3, elocalcitol	UUI/OAB	No urodynamic effect
Calcium antagonists	OAB	No effect
K^+ channel openers (ZD0947, flupirtine)	OAB	Dose capped by heart; flupirtine hepatotoxic
URO-902 (BK channel gene therapy)	OAB	Phase 1 only
RimabotulinumtoxinB	Toxin A-refractory DO	Short duration (≤3 months)
Intraprostatic onabotA, PRX302, NX-1207	BPH LUTS	= sham injection

9 · Landmark trials

Trial	Design	Result	Take-home
SYNERGY / II	Phase III, 12 weeks / 12 months	Solifenacin 5 + mirabegron 50 > monotherapy for UI	Effective, safe long-term
BESIDE	Solifenacin 5 non-responders	Add-on > solifenacin 5, = solifenacin 10	Alternative to dose escalation
MILAI II	52 weeks, mirabegron non-responders	Four antimuscarinic add-ons safe, effective	Add antimuscarinic to mirabegron
DIGNITY	Pooled RCTs, NDO	OnabotA 200 U equal in anticholinergic users/non-users	Efficacy independent of anticholinergic
EMBARK	RCTs, refractory OAB	OnabotA 100 U effective	Standard idiopathic OAB dose

115 Electrical Stimulation and Neuromodulation in Storage and Emptying Failure

- P4 P3 Neuromodulation for voiding dysfunction, including sacral neuromodulation; Sacral neuromodulation — *afferent/forebrain mechanism, FDA indications table, PNE vs staged lead, S2/S3/S4 responses, ≥50% test, InSite, ROSETTA, PTNS technique, SUmiT, TTNS, implantable tibial, pudendal, DGNS, IVES, Brindley*

1 · Evolution and mechanism

- Direct detrusor/sphincter stimulation failed (painful, local, bladder and sphincter contracted together) → **indirect via afferents**.
- Milestones MCQ: SNM ~1985 (Tanagho, Schmidt); first tibial nerve stimulation 1983 (McGuire); PTNS from acupuncture point **SP6**; first tibial implant 2006.
- **Exclude anatomical causes first** (stones, cancer, fistula, poor compliance, stricture, prostate, prolapse).
- **Current theory**: sacral roots mixed → afferents → forebrain awareness/alertness areas → less PMC firing in OAB; in underactive bladder raised filling awareness lets PMC switch on.
- Normal voiding: EUS relaxes, detrusor contracts 1–2 s later. Old efferent theory implausible: benefit needs no pelvic floor contraction; purely sensory nerves work equally.
- **Paradox**: in retention SNM works mainly by **striated sphincter relaxation**, not detrusor contraction.

2 · Sacral neuromodulation

- **Third-line** for idiopathic OAB (AUA/SUFU = EAU). Pre-treatment: frequency–volume diary (OAB definition ignores excess intake); exclude UTI, significant PVR, stones, cancer. Urodynamic DO **does not predict** success.
- **Non-obstructive urinary retention (NOUR)**: PVR is the only objective outcome; Fowler syndrome predicts response; age lowers odds.

FDA approved	Not approved
UUI	Bladder pain syndrome
Urgency–frequency	Pelvic pain syndrome
Chronic NOUR	Neurogenic bladder

- Bladder pain: offer selectively if coexisting OAB. Neurogenic bladder unapproved but pooled success comparable. †Faecal incontinence approved.
- Test success: ≥**50%** improvement in ≥1 diary parameter vs 3–5-day baseline; retention — catheterisation ↓ ≥50%, ideally none.
- **S3 target** (≈9 cm above coccyx tip, 1–2 cm lateral): anal "bellows" + **great-toe flexion**; S2 calf/foot rotation; S4 perineal only.
- PNE: office, temporary lead, 5–7 days; staged: permanent tined quadripolar lead under fluoroscopy, test up to 4 weeks, generator in buttock.
- PNE: lead migrates → false-negative; lower conversion; not used in NOUR. Staged: stage 2 at 2–4 weeks; failure → remove lead. Aim: responses on all four contacts at low amplitude. Prophylaxis cefazolin + metronidazole (± MRSA cover).

- Programming: 210 µs, 10–16 Hz, sensory threshold; reprogramming ≈3 times in year 1; cycling = continuous; **bilateral not superior** (RCT).
- Battery: non-rechargeable ≥**10 years**, rechargeable ≥15 years. **MRI**-conditional whole-body at **1.5 and 3 T**; at replacement remove all old components.
- Complications: revision ~3%, explantation ~9% at 2 years (ROSETTA); †site pain, leg pain → reprogram, then revise; infection → explant.

3 · Tibial nerve stimulation

ALGORITHM

Supine → 34-gauge needle ≈3–4 cm above medial malleolus, behind tibia, at 60–90° → ground pad same leg → 20 Hz, 200 µs, 0–10 mA → confirm **great-toe flexion/toe fanning** or sole tingling → **12 weekly sessions × 30 min** → taper → maintenance

- Mechanism also cortical reorganisation. Indications: drug-refractory/intolerant OAB; safe in neurogenic OAB (MS, PD, stroke); children failing bladder training and drugs.
- **TTNS**: home surface electrodes; skin impedance → less effective than PTNS, cheaper. Vs tolterodine: equal efficacy, fewer side effects.
- Success ≥50% fall in urgency/UUI; **relapse likely without maintenance**. Better without DO; poor mental-health score predicts failure. PTNS response does not guarantee implant response.

4 · Other targets

- **Pudendal** (S2–S4, Alcock canal): posterior approach (perineal more infection); better than SNM in blinded crossover; salvage after failed SNM; not FDA approved.
- **Dorsal genital nerve**: limited by electrode site. **Sacral dermatome** (transcutaneous): ↑ capacity; cannot abort contractions. **Intravaginal** for SUI: safe, low-quality evidence.
- †**Intravesical stimulation**: neurogenic underactive bladder, incomplete lesions; afferent pathway must be intact; evidence limited.

5 · Sacral anterior root stimulation (Brindley)

- Anterior roots → voiding, defecation, erection in complete SCI; with **posterior sacral rhizotomy**: abolishes reflex incontinence and autonomic dysreflexia, ↑ capacity/compliance, protects kidneys.
- †"Post-stimulus voiding" (sphincter relaxes faster than detrusor). †Rhizotomy cost: reflex erection, reflex ejaculation, residual sacral sensation lost.

6 · Landmark trials

Trial	Design	Result	Take-home
SUmiT	RCT, PTNS vs validated sham, 12 weeks, n = 220	Responders 55% vs 21%	First sham-proven PTNS efficacy; maintenance keeps 77% relapse-free at 3 years
InSite	RCT, SNM vs standard medical therapy, mild–moderate OAB	Success 61% vs 42% at 6 months	SNM beats continued drugs

Trial	Design	Result	Take-home
ROSETTA	RCT, onabotulinumtoxinA 200 U vs SNM, refractory UUI women	Toxin slightly fewer leaks at 6 months; equal at 2 years; more UTI, some CIC	Both valid third-line; registered OAB toxin dose is 100 U

★ Putting it together: voiding dysfunction decisions

A · Functional classification as treatment map

	Bladder	Outlet
Failure to store	Overactivity/low compliance → antimuscarinic, β3-agonist, onabotulinumtoxinA, neuromodulation, augmentation	Sphincter weakness → PFMT, duloxetine (Europe), sling, AUS, bulking
Failure to empty	Underactive/acontractile → CIC; SNM for non-obstructive retention	Obstruction (anatomical, dyssynergia) → α-blocker, prostate/bladder-neck surgery, sphincterotomy, CIC

Neither fixable → circumvent: CIC, indwelling catheter, diversion (*→ Ch 104*).

B · Lesion level and urodynamics

Lesion	Detrusor	Striated sphincter	Key risk
Above pons (stroke, PD, dementia)	Overactive	Synergic	UUI; low upper-tract risk
Pons–sacral cord (SCI, MS)	Overactive	**Dyssynergic**	High pressure → upper tracts; AD if above T6
Sacral/cauda equina/peripheral (diabetes, pelvic surgery)	Areflexic, ↓ sensation	Fixed tone	Retention, overflow; compliance may fall

KEY POINT

Dyssynergia only with lesions between pons and sacral cord. Leak point/storage pressure ≥40 cmH_2O or poor compliance threatens kidneys — treat (*→ Ch 107, 109*).

C · OAB ladder

ALGORITHM

1. Behavioural: fluids/caffeine, bladder training, PFMT, constipation, drug review.

2. Antimuscarinic or β3-agonist (β3/low-CNS in elderly); combine if partial; PVR in obstructed men (*→ Ch 113*).

3. OnabotulinumtoxinA (100 U idiopathic, 200 U neurogenic; warn CIC), PTNS, or SNM (staged test, ≥50% improvement) (*→ Ch 115*).

4. Augmentation or diversion.

D · SCI bladder

ALGORITHM

Spinal shock: catheter until reflexes return.

Then UDS; CIC + antimuscarinic for high-pressure NDO; toxin; sphincterotomy/diversion if CIC impossible; lifelong upper-tract surveillance.

AD (above T6): hypertension, bradycardia, headache, flushing → sit up, loosen clothing, remove stimulus (catheter, bladder, bowel), short-acting antihypertensive (*→ Ch 109*).